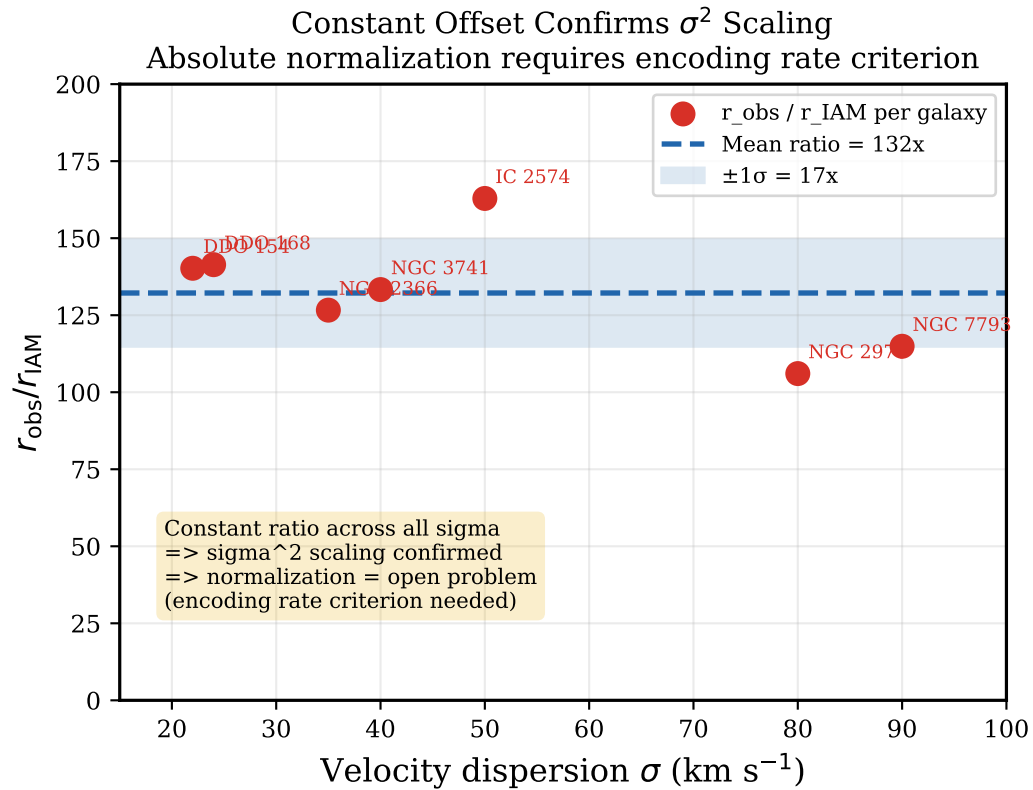
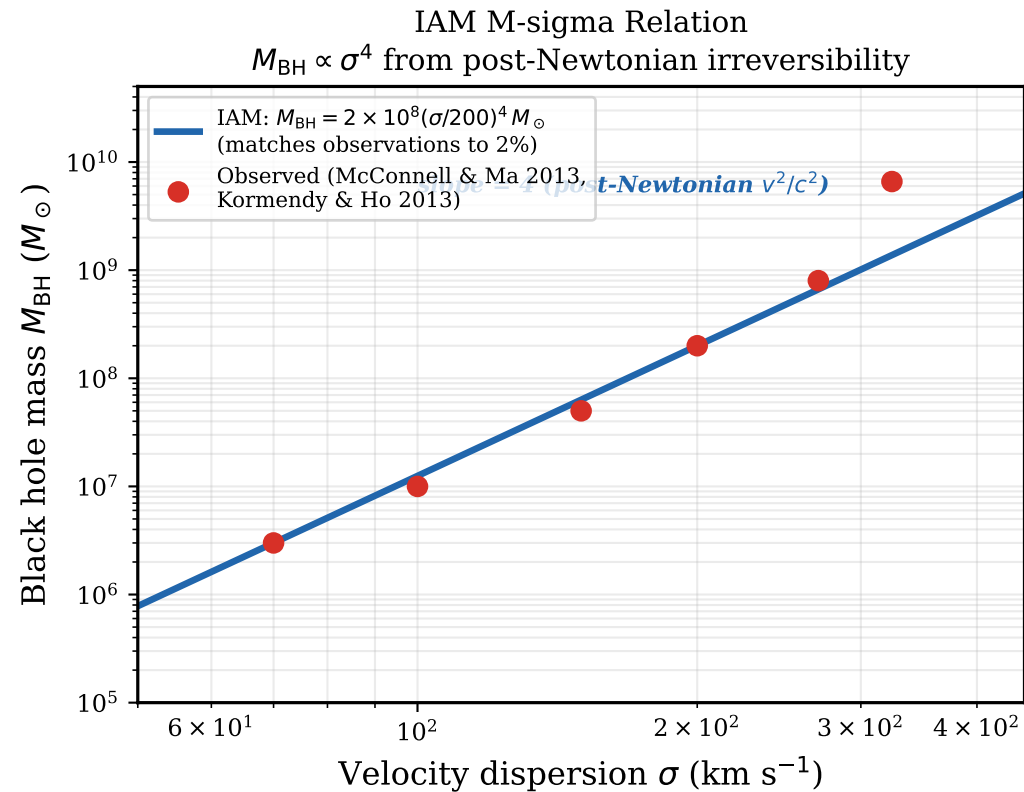
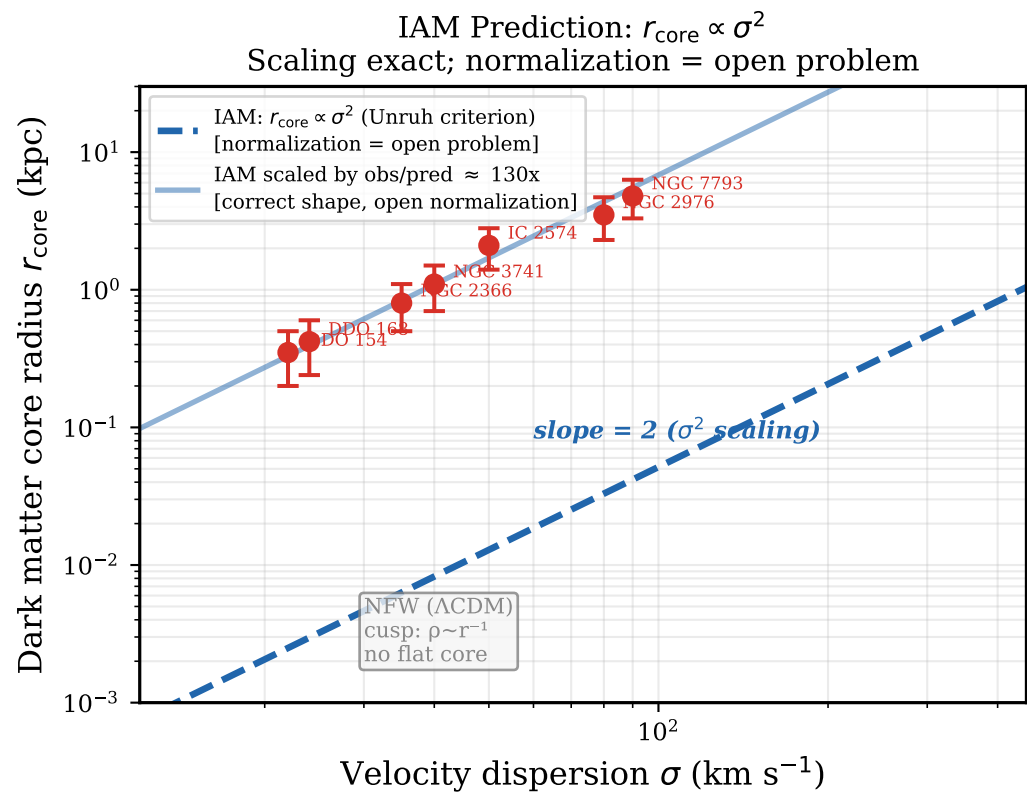


IAM Prediction: Dark Matter Core Radius Scaling

$r_{\text{core}} \propto \sigma^2$ from Black Hole Encoding Dominance (zero free parameters)



IAM Cusp–Core Resolution (Timestamped Prediction, March 16, 2026)

Why black holes create cores :

BH = local encoding surface (not just mass)
Forms when decoherence rate saturates
Bekenstein-Hawking bound [BH Thermo paper]

Inside r_{core} (BH encoding zone) :

Decoherence kinetic half -> BH horizon
Kinetic energy dissipates as THERMAL
Hawking radiation (heat, not structure)
Geometric half deposited at LARGER radii
=> NO dark matter at center => CORE

Outside r_{core} :

Cosmic horizon = dominant encoder
Normal virial partition -> dark matter
=> standard density profile => NFW-like

Prediction (zero free parameters) :

$$r_{\text{core}} = \sqrt{GM_{\text{BH}}/(cH_0)}$$

$$M_{\text{BH}} = 2 \times 10^8 (\sigma/200)^4 M_{\odot}$$

$$\Rightarrow r_{\text{core}} \propto \sigma^2 \text{ (EXACT)}$$

Status :

σ^2 scaling: DERIVED, exact
Absolute normalization: OPEN PROBLEM
(encoding rate criterion not yet derived)
Systematic offset: $\sim 130x$ (constant)

Mahaffey 2026 (BH Thermo, M-sigma, BH Info Paradox, Theory Papers)
McConnell & Ma 2013, ApJ 764, 184
Kormendy & Ho 2013, ARA&A 51, 511
Oh et al. 2015, AJ 149, 180
de Blok et al. 2008, AJ 136, 2648
Navarro et al. 1997, ApJ 490, 493
Planck 2020, A&A 641, A6